BLACKBOX

THIRDUNI • OCTOBER 2026

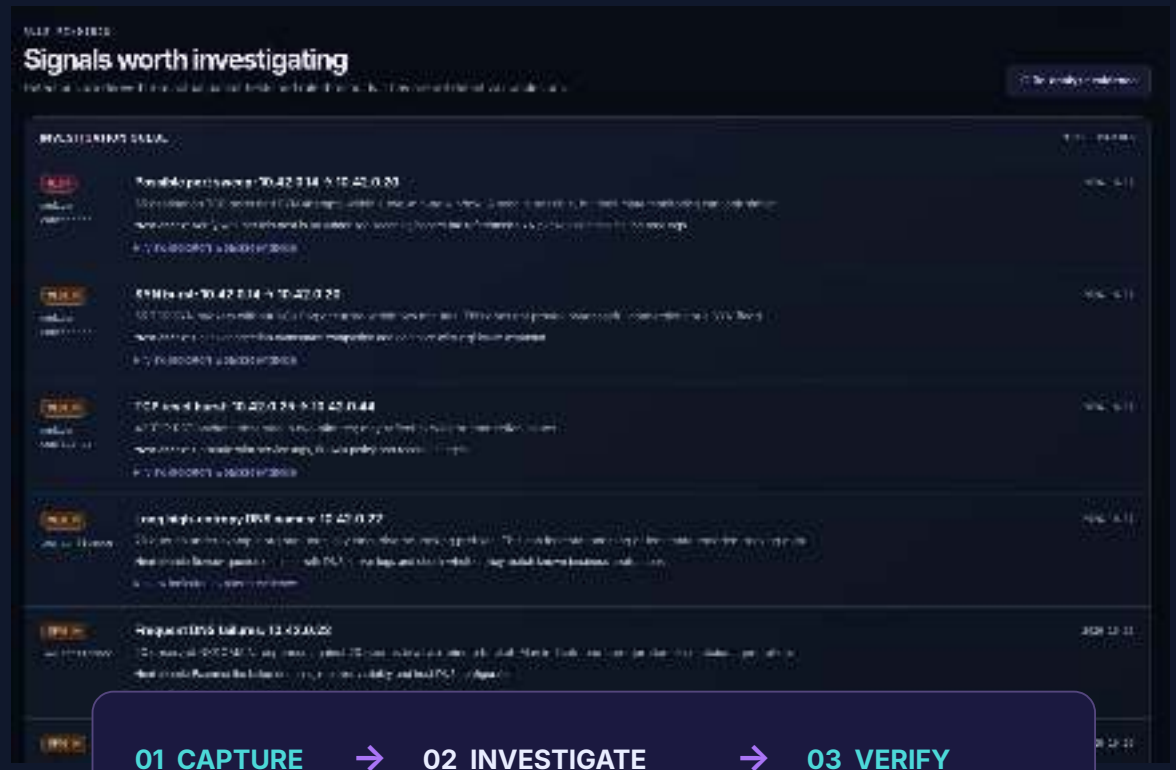
FROM PACKETS TO PROOF.

A network incident investigator that turns captured traffic into traceable questions, findings and evidence.

REAL PCAP INPUT

TRACEABLE FINDINGS

READ-ONLY COPILOT



Security evidence is everywhere. Clarity is not.

Packet captures are rich in detail, but investigation still means connecting many small observations.

THE ANALYST'S CHALLENGE

Thousands of packets.
Scattered clues.
Too much manual correlation.

The problem is not collecting packets — it is explaining what the traffic supports and what remains uncertain.

1 **SEE THE TRAFFIC**
Which devices communicated, when and using what protocol?

2 **SURFACE SIGNALS**
Which patterns deserve further investigation?

3 **TRACE THE EVIDENCE**
Which packet references support every claim?

BLACKBOX helps analysts investigate signals — it does not automatically confirm a cyberattack.

One capture. A traceable investigation.

Each stage uses facts derived from the selected file and keeps a route back to the source evidence.

01
IMPORT
PCAP / PCAPNG
Input capture

02
PARSE
Packets + timestamps
Structured records

03
ANALYZE
Flows + heuristics
Review signals

04
INVESTIGATE
Timeline + AI phrasing
Context & evidence

05
REPORT
Markdown / JSON
Shareable case

CONTROLLED EVIDENCE

Local processing first. Cloud AI is optional and receives selected metadata, not raw PCAP.

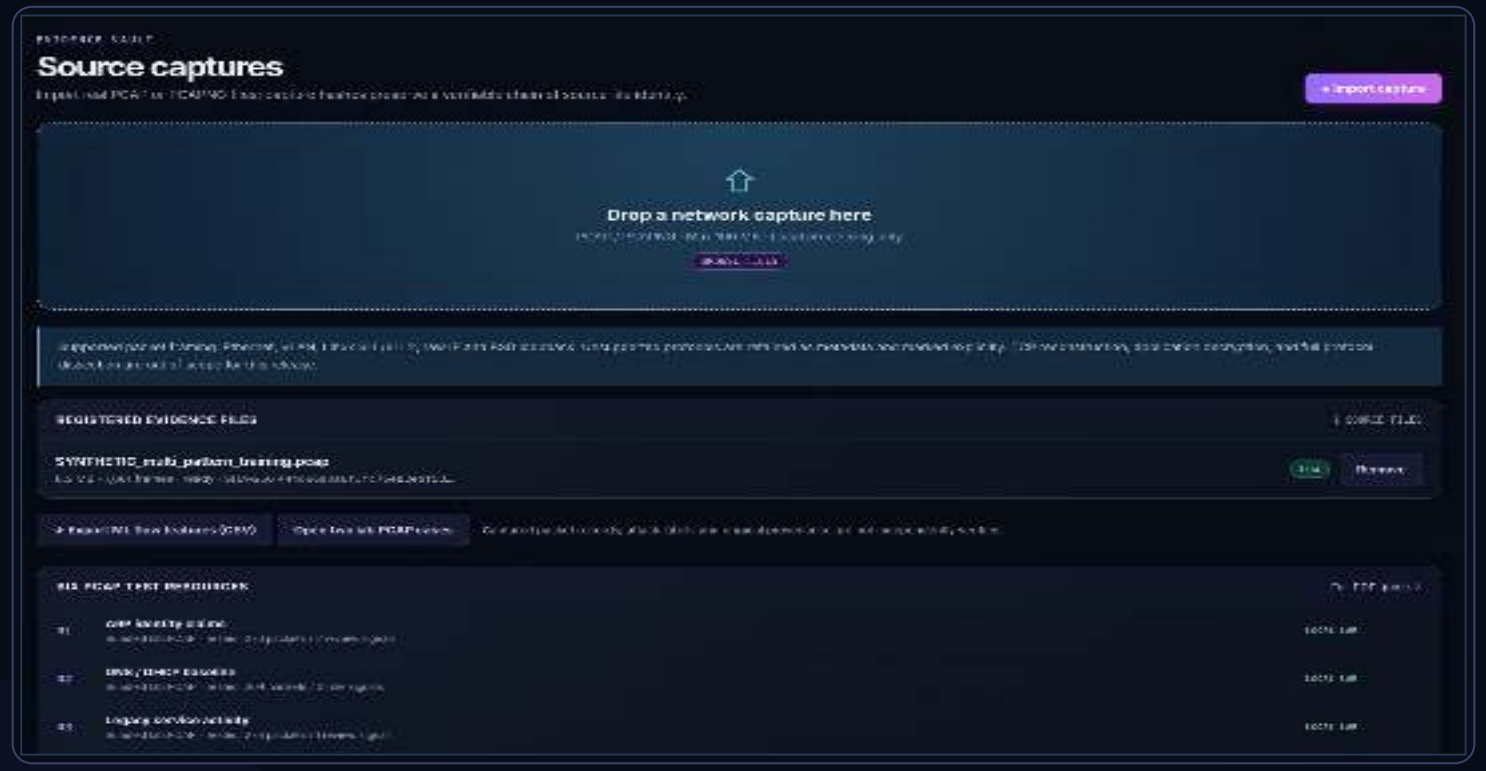
ML EXTENSION

Flow-feature CSV export works; a trained ML detection model is not yet integrated.

Key distinction: rules and evidence work now; a future ML model may add a separate prediction layer.

Start with the source. See the case.

Users can bring a capture into a case, verify file identity, and orient themselves from the overview.



01 **Create a case**

Keep unrelated investigations separate.

02 **Import PCAP / PCAPNG**

Upload a capture collected with authorization.

03 **Check the evidence**

Confirm file, frame count, status and SHA-256.

TRY IT → Evidence Vault / Import capture / Open Overview • Local app: 127.0.0.1:8765

An alert is a lead — not a verdict.

BLACKBOX links rule hits to the packet observations that triggered them.



SYNTHETIC TRAINING EXAMPLE

Possible port sweep

10.42.0.14 → 10.42.0.20

55 destination TCP ports and repeated SYN attempts within a two-minute window.

CHALLENGE & VERIFY

Use BLACKBOX TRACE to test alternative explanations; verify linked packets and host logs.

Packet IDs + source capture identity remain linked to each reported finding.

Who communicated — and when?

Two complementary views make the network easier to investigate without inventing causal explanations.



NETWORK MAP Observed IP-to-IP relationships, not physical locations.

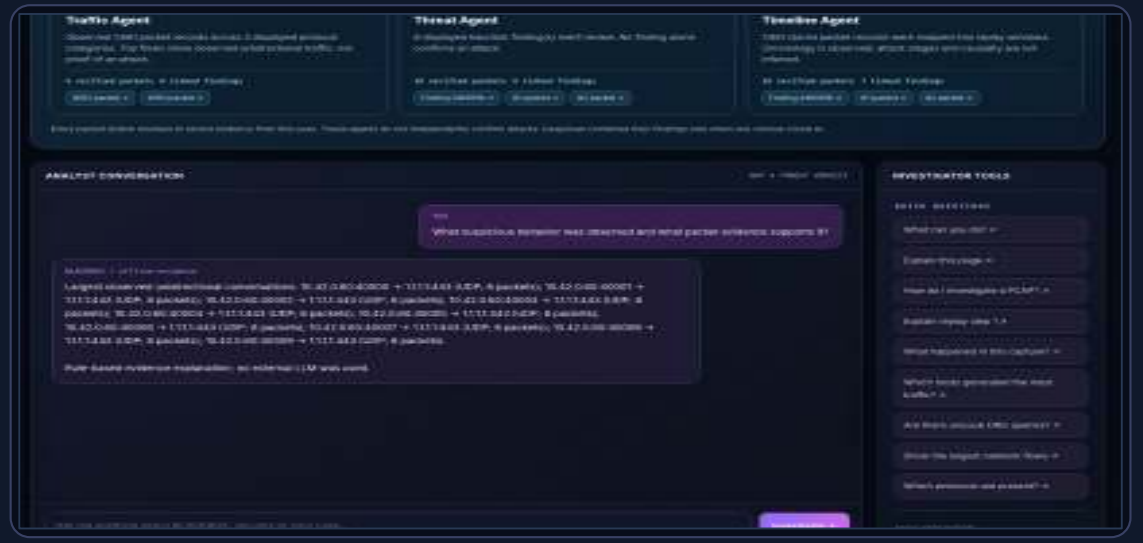


INCIDENT REPLAY Step through captured timestamps and linked findings.

A timeline shows sequence, not intent. Confirm causality with independent evidence.

Ask, inspect, document.

The Investigator explains selected case evidence. The final report preserves observations and limitations.



INVESTIGATOR Ask which device, what evidence, and what remains uncertain.



CASE REPORT Export Markdown or JSON for further analyst review.

TRACE = competing explanations + packet evidence. LLM = optional explanation. Trained ML = pending.

Real workflow. Transparent results.

The release includes synthetic test captures; live deployment and evaluated ML are future work.

CAPTURE	PACKETS	RULE HITS
Multi-pattern stress*	**7,861**	**9**
TCP sweep demo	31	2
ARP conflict demo	9	1
DNS / DHCP demo	39	0

*Stress capture generated locally. Every included demo is synthetic; not proof of real attacks.

HOW TO TRY IT

01 Extract ZIP and run start_windows.bat

02 Open 127.0.0.1:8765

03 Import a permitted PCAP in Evidence Vault

04 Open Findings → Packet Explorer

05 Ask Investigator → export report

SOLO CREATOR Ikram Salhi — Network & Embedded Systems

Architecture · code · testing · presentation

NEXT: launch a read-only public demo; independently evaluate any future ML detector.